

Heart Disease Prediction Report

Supervised Machine Learning Classification using Logistic Regression

Machine Learning

Healthcare Analytics

Binary Classification

PROJECT OVERVIEW

Heart disease remains one of the leading causes of global mortality. Early, accurate detection plays a pivotal role in enabling preventative care and optimizing clinical outcomes. This document details a supervised machine learning methodology designed to evaluate patient physiological parameters and assess the likelihood of heart disease presence using a trained **Logistic Regression** model.

DATASET OVERVIEW

The classification framework utilizes the standard Heart Disease Dataset, encompassing comprehensive demographic, physical, and clinical diagnostics gathered from patient examinations.

TOTAL INSTANCES

1,025

TOTAL FEATURES

14 Columns

MISSING VALUES

0 (Clean)

TECHNICAL STACK

Language: Python

Data Processing: Pandas, NumPy

Model Suite: Scikit-Learn

MACHINE LEARNING WORKFLOW

- Data Ingestion & Integrity Check:** Load dataset, perform structural verification, confirm zero null values across all attributes, and evaluate summary statistics.
- Feature Matrix Construction:** Separate independent physiological variables (\$X\$) from the binary diagnosis label (\$Y\$).
- Stratified Data Partitioning:** Perform an 80/20 train-test split utilizing target stratification to ensure both sets retain identical target class distributions.
- Model Fitting & Optimization:** Fit a binary Logistic Regression classifier using optimization algorithms suited for convex loss functions.
- Performance Evaluation:** Compute accuracy scores across both training and unseen testing partitions to measure generalization capacity and prevent overfitting.

FEATURE ATTRIBUTE DEFINITIONS

The 14 clinical features analyzed by the classification pipeline are defined as follows:

Attribute	Description	Values / Scale
age	Age of the patient in years	Continuous (e.g., 29 to 77)
sex	Biological gender	1 = Male; 0 = Female
cp	Chest pain type experienced	Values 0–3 (Typical, Atypical, Non-anginal, Asymptomatic)
trestbps	Resting blood pressure on admission	Continuous (mm Hg)
chol	Serum cholestorol level	Continuous (mg/dl)
fbss	Fasting blood sugar > 120 mg/dl	1 = True; 0 = False
restecg	Resting electrocardiographic results	Values 0, 1, 2 (Normal, ST-T wave abnormality, LV hypertrophy)
thalach	Maximum heart rate achieved	Continuous (bpm)
exang	Exercise induced angina	1 = Yes; 0 = No
oldpeak	ST depression induced by exercise	Continuous (Relative to rest)
slope	Slope of the peak exercise ST segment	Values 0–2 (Upsloping, Flat, Downsloping)
ca	Number of major vessels colored by fluoroscopy	Values 0–3
thal	Thalassemia blood disorder test	0 = Normal; 1 = Fixed defect; 2 = Reversible defect
target	Diagnosis outcome label	1 = Heart Disease Detected; 0 = Normal / Healthy

MACHINE LEARNING WORKFLOW

- Data Ingestion & Integrity Check:** Load dataset, perform structural verification, confirm zero null values across all attributes, and evaluate summary statistics.
- Feature Matrix Construction:** Separate independent physiological variables (\$X\$) from the binary diagnosis label (\$Y\$).
- Stratified Data Partitioning:** Perform an 80/20 train-test split utilizing target stratification to ensure both sets retain identical target class distributions.
- Model Fitting & Optimization:** Fit a binary Logistic Regression classifier using optimization algorithms suited for convex loss functions.
- Performance Evaluation:** Compute accuracy scores across both training and unseen testing partitions to measure generalization capacity and prevent overfitting.